



UNIVERSITY OF SOUTHERN MAINE

Information Retrieval, Fall 2022, Project 1

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Due: October 6, 2022

Notes for submission:

1. Submit your file(s) with the correct naming as IR_Project1_StudentName.
2. Two files can be uploaded, one .zip file having all the codes (directory is zipped and it is named codes) and one .pdf file. There would be a penalty for uploading wrong formatted files. Any other formatting will be ignored.
3. Codes should be well-structured with comments to run.
4. Before **October 18**, codes should be available on your GitHub Repo. Failure to have codes publicly available results in a 20% reduction in grade.
5. Any assumptions made by students should be explicitly mentioned in the submitted document.
6. Answers to the questions should be easy to detect. Both codes and documents should refer to the id of the question/sub-question being answered.
7. The template for your reports can be found here:

<https://www.overleaf.com/latex/templates/association-for-computing-machinery-acm-sig-proceedings-template/bmvfhcdnxfty>

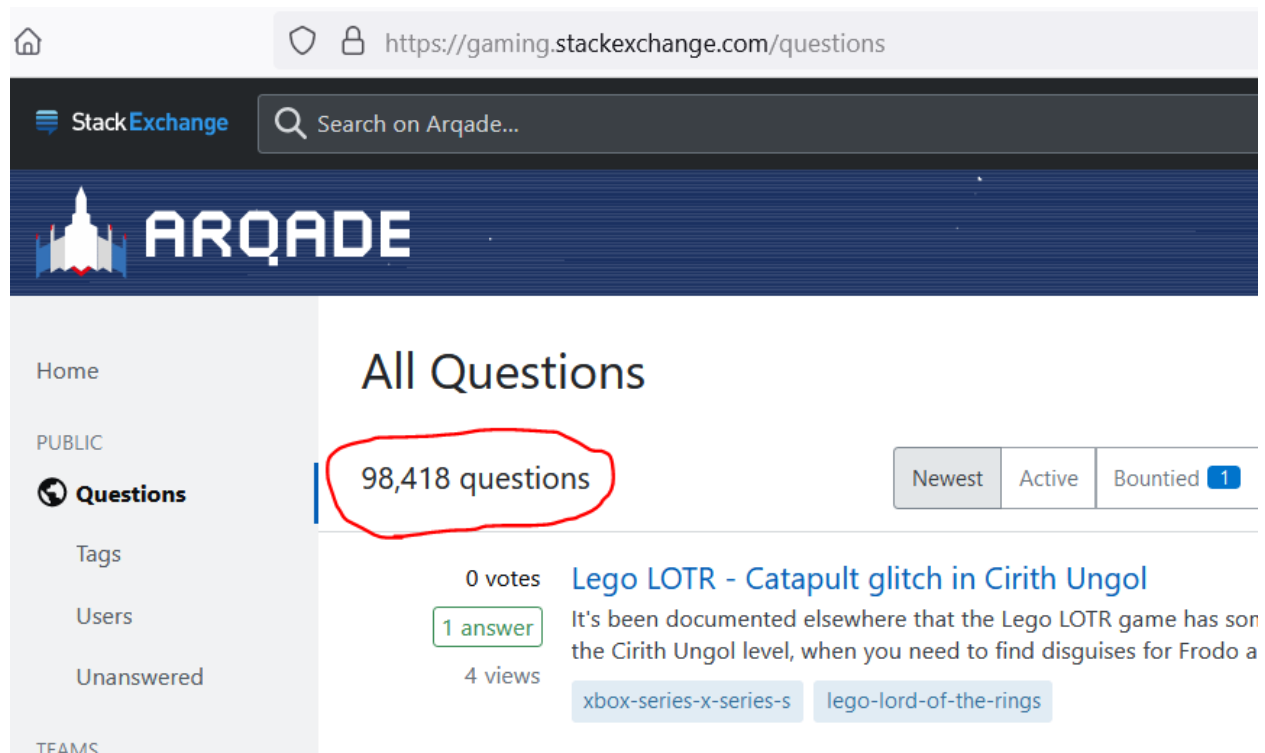
Part 1

Here the steps for the project Part 1 are described.

Step 1. Choosing StackExchange Topic

StackExchange has a variety of sites. You can view them [here](#). The Internet Archive provides a list of archived StackExchange websites [here](#). All the websites here are in the same format as Coffee StackExchange that we use for our assignments.

You are free to choose any of these sites for your project. However, there is only one hard constraint: the website should have more than 50K questions posted. You can easily check this by going to the website and clicking on the Questions. For example, for the gaming StackExchange, go to <https://gaming.stackexchange.com/> and then click on Questions as shown below:



Another way of testing this is to download the post file and read the number of posts, as practiced in previous assignments. Note that **you cannot change the topic** in the second and third parts. Therefore, you may be careful about your choice. You have to create a judgment file for your queries (explained in the next sections), therefore you may choose a topic that you are familiar with. Also, some of the websites may bring you unseen difficulties. For example, if you choose MathStackExchange, not only you have to handle text, but you should find a way to handle formulas, which are usually represented in a different format than text.

Step 2. Preparing the Files

After selecting the topic, the next step is to prepare your files. Let's continue with the gaming website example. Go to the Archive [link](#) and download the corresponding .7z file for the topic you choose. After downloading the file, you will see different .xml files as shown below:

File folder					
Badges.xml	48,747,199	4,167,832	XML Document	6/5/2022 10:14 ...	E3C9EAEB
Comments.xml	99,050,309	20,145,293	XML Document	6/5/2022 10:14 ...	D7743305
PostHistory.xml	531,069,306	108,391,883	XML Document	6/5/2022 10:14 ...	9C8B3603
PostLinks.xml	3,904,286	449,549	XML Document	6/5/2022 10:15 ...	0BB32401
Posts.xml	285,377,816	57,833,189	XML Document	6/5/2022 10:15 ...	E1187709
Tags.xml	465,842	11,802,990	XML Document	6/5/2022 10:15 ...	2D43FF0C
Users.xml	68,058,720	?	XML Document	6/5/2022 10:15 ...	7E5D8032
Votes.xml	156,593,220	7,921,293	XML Document	6/5/2022 10:15 ...	E6F036D6

There are other files but the main file that we use for searching in the project is the Posts.xml file that has the questions and answers on the website you chose. Extract this file and check its content.

```
<?xml version="1.0" encoding="utf-8"?>
<posts>
  <row Id="1" PostTypeId="1" AcceptedAnswerId="26" CreationDate="2010-07-07T19:06:25.043" Score="49" ViewCount="5816" Body="<p>Now that the Engineer update has come, th</p></row>
  <row Id="2" PostTypeId="1" AcceptedAnswerId="184" CreationDate="2010-07-07T19:07:58.427" Score="15" ViewCount="11336" Body="<p>I know I can create a Warp Gate and tel</p></row>
  <row Id="3" PostTypeId="1" AcceptedAnswerId="56" CreationDate="2010-07-07T19:09:46.317" Score="19" ViewCount="4805" Body="<p>Steam won't let me have two instances runn</p></row>
  <row Id="4" PostTypeId="1" AcceptedAnswerId="14" CreationDate="2010-07-07T19:11:05.640" Score="33" ViewCount="4785" Body="<p>When I get to the insult sword-fighting s</p></row>
```

No matter what topic you have chosen, the Posts.xml files are similar and can be read with the code used for your assignments. Each row in this file represents a post (answer or question). Each post has different attributes, including PostTypeId (1 for question, 2 for answer), CreateDate, and Score.

Important Note: For the implementation of all your codes, it is strongly recommended that you start with small samples and check your code manually for correction. For that, you can easily choose a few first lines of your Posts.xml file and save a .xml file as Posts_small_test.xml and check your codes on that file first.

Each post (question/answer) has a unique Id. For your analysis, you may need to view the post. To do that, one possible way instead of directly using the XML files is to use the website you have chosen. For example, to view the post with id '8000' we would use this link:

<https://gaming.stackexchange.com/questions/8000>

Step 3: Analyze the Collection (20%)

In Information Retrieval, to develop a search system, we first analyze the collection to inform ourselves about the characteristics of the data. In this step, the goal is to get a sense of what the website you have chosen looks like. For that answer the following questions:

- 1- What are the common terms used in the collection? Generate a **Word Cloud** for the top-20 frequent words. Then remove the stopwords using NLTK and regenerate the Word Cloud to the top-20 frequent words. Compare the differences. Plot the frequency of words and discuss if they follow Zipf's law.
- 2- What are the top-10 common question tags used for the questions? For the top-20 tags for questions, show their distribution. There should be another category for other tags.
- 3- What is the average number of words and sentences in question (body and title) and answers?
- 4- What is the average number of answers given to each question? How many questions have no answer? How many questions have an accepted answer?
- 5- Select examples of questions with no answer. Manually or automatically check these questions and discuss what are the possible reasons that these questions are not answered. For example, can it be the case that questions with no answers are longer than those with answers?
- 6- How many of the accepted answers are the first answers that are posted? Is there a specific pattern in the reputation score of the person who answered the question? (Use correlation measures for this) Are always the accepted answers the ones with the highest score? If not, provide an example.
- 7- For questions with no answer, measure the question readability. Is there a correlation between the readability of the question and the chance of receiving an answer?
- 8- How many of the questions have duplicate questions? Normalize the text and then select a question with its duplicate and report the number of common terms in their title and body.
- 9- Choose 5 questions with comments. Discuss what is happening in the comments. Provide an example for your discussion. For example, is the asker clarifying the question? Is the asker adding more details to the questions?
- 10- Analysis of your choice! In this part, you should do one more analysis that you find interesting about your collection. To do this you may do further reading on papers such as: *"Question Quality in Community Question Answering Forums: A Survey"* or *"Assessing the quality of answers autonomously in community question-answering"*.

Step 3: Boolean Search System (25%)

In this step, you will develop a boolean search system. Your search system should have the ability to search both with “OR” and “AND” operators. Your search system will return the top 50 answers for each input query with their post ids. Tokenize your text and apply any preprocessing step needed. Note that your files are XML, and tags such as <p> should not be indexed. In your report, mention how big your inverted index is and how much time did take to build the inverted index (specifying your machine setting).

Step 4: Simple Inverted Index Retrieval (25%)

Create an inverted index with term frequency for each term. For search, use the term-at-a-time approach. Similar to a boolean search, your system returns top-50 answers. In your report, mention how big your inverted index is and how much time did take to build the inverted index (specifying your machine setting).

Step 5: Building Query Set and Evaluating the Results (35%)

After preparing your search systems, it is time to evaluate them! You will consider a set of 20 queries. You will explore the questions you have in your selected Stack Exchange, then select 20 questions and use their titles as the query. Make sure that your queries are diverse.

You can consider the Border’s categories. You may also consider different lengths for your queries (Short, Medium, Long) or question tags (having questions from different tags). For each query, assess the top-10 results with the relevance scores of Relevant, Partially Relevant, and Not Relevant. For **one particular query**, do a dual-assessment and report your agreement measure. (If you are doing the project yourself, ask a friend to do it for you)

To decide the relevance, you can use the question body to gain information about the information need and based on the information need, decide the relevance.

For each system, report the average retrieval time, the Precision, and nDCG values at cuts of 5 and 10. Then discuss the strength and weaknesses of each model.

Report

In your report, you will have the following sections:

Introduction: Introducing the website you have chosen and the reasoning of your choice (Step 1)

Test Collection: You will answer the questions in Step 3

- **Collection:** Analysis of your collection
- **Queries:** You will explain what queries you have chosen here. (Step 5) You will discuss your choices and the categories

Boolean Retrieval: You will describe your system with a link to your GitHub repo. You will include the answers to questions in Step 5

Simple Inverted Index: You will describe your system with a link to the same GitHub repo. You will answer the questions in Step 5 in this section

Results: In this section, you will compare the results of two systems. Provide examples for which one system did better than the other and discuss why.

Conclusion: You will provide 2-4 paragraphs here to conclude your main findings and experience with the project

References: You can use any references or existing codes. However, you should not miss citing and referring to any resources you have used. Missing references can lead to a plagiarism penalty.

For your report use this template:

<https://www.overleaf.com/latex/templates/association-for-computing-machinery-acm-sig-proceedings-template/bmvfhcdnxfty>

A cleaner version of this is available on the course website.

Presentation

Each group will have 12-15 minutes to present their work. During this time, each group member will present their contribution to the project. In the presentation, the team should run their systems online with an example query and show the results.